

WASP-80b

R-2767

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-80_260730 · created 2026-09-03T22:51:03+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461251.7984 +/- 0.0063 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.228 +/- 0.068
Transit depth (Rp/Rs)^2	5.18 +/- 3.07 [%]
Orbital Inclination (inc)	87.19 +/- 1.69
Airmass coefficient 1 (a1)	46.82 +/- 23.94
Airmass coefficient 2 (a2)	-2.96 +/- 0.84
Scatter in the residuals of the lightcurve fit is	55.81 %
Best Comparison Star	None
Optimal Aperture	3.16
Optimal Annulus	8.54
Transit Duration (day)	0.073 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.228 ± 0.068 vs 0.1714 (deviation 0.42σ)
Duration fit vs published	0.073 d vs 0.0892 d (deviation 0.36σ)
Mid-time O–C	-1.3 min
Depth SNR	1.7
Residual scatter	55.81 %

Light curve

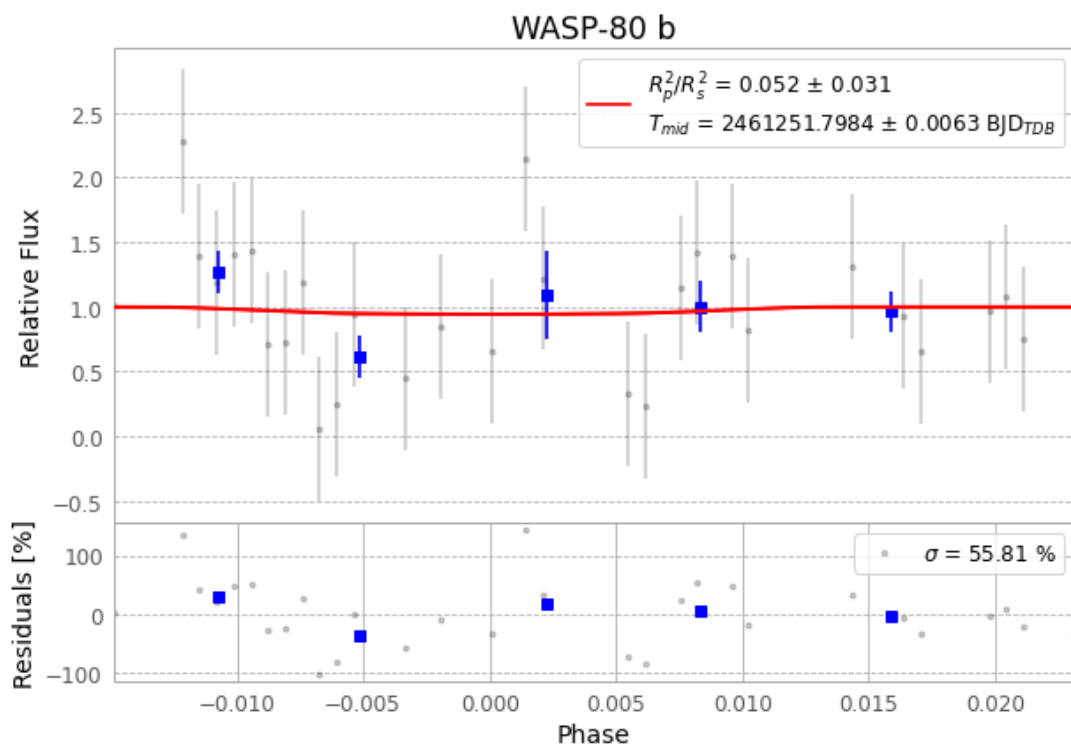


Figure 1 — FinalLightCurve_WASP-80 b_2026-07-29.png (SHA-256 04aa00302066871f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [521, 111], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 12c9ec6ca5397d4d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), WASP-80260730054815.FITS → WASP-80260730090615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-80-260730.log (95754460ea54...), FinalLightCurve_WASP-80 b_2026-07-29.png (04aa00302066...), FinalLightCurve_WASP-80 b_2026-07-29.csv (565c1eb487cf...)

Compute resources

Wall time 191.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 22:51Z.